Simultaneous Equations

Name _____ Date _____ No calculator Version 1

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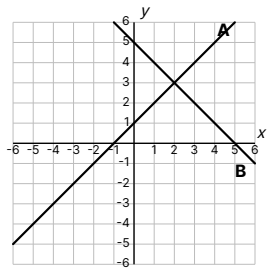
KEY WORDS simultaneous equations *(whārite pāhekoheko)* · elimination · substitution *(whakakapinga)*

A The solution on a graph ●○○

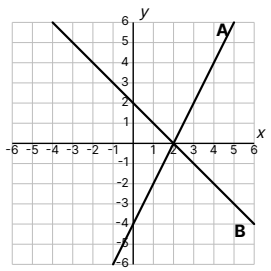
Remember: Each equation is a straight line. The solution of the pair is the one point that is on both lines: where they cross. Read its coordinates, and check them in both equations.

1. Lines A and B are drawn on each grid. Write down the solution of the pair of equations.

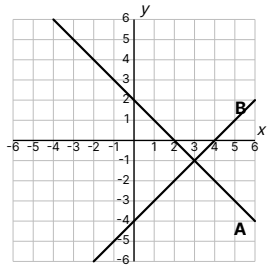
a) A: $y = x + 1$, B: $y = -x + 5$



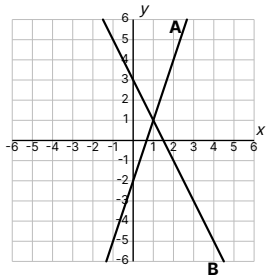
b) A: $y = 2x - 4$, B: $y = -x + 2$



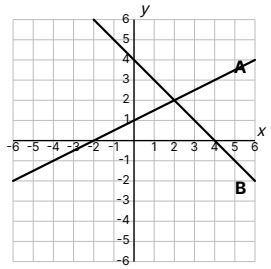
c) A: $x + y = 2$, B: $y = x - 4$



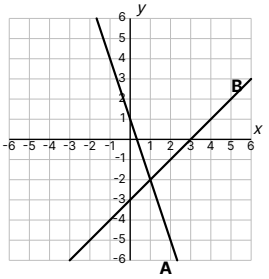
d) A: $y = 3x - 2$, B: $y = -2x + 3$



e) A: $y = \frac{1}{2}x + 1$, B: $x + y = 4$



f) A: $y = -3x + 1$, B: $y = x - 3$



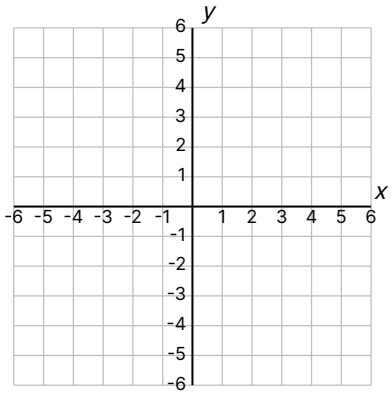
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B Solving by drawing ●●○

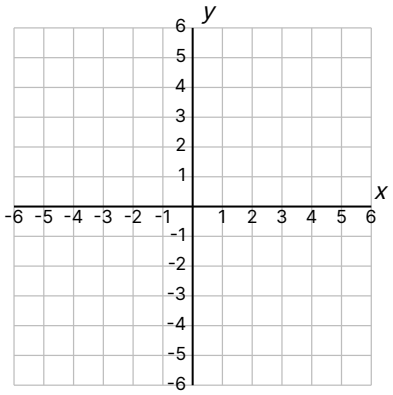
Remember: Draw each line from its gradient and intercept (or from a table), rule it right across the grid, and read the point where the two lines cross. Label it P.

2. Draw both lines on the grid and write down the solution.

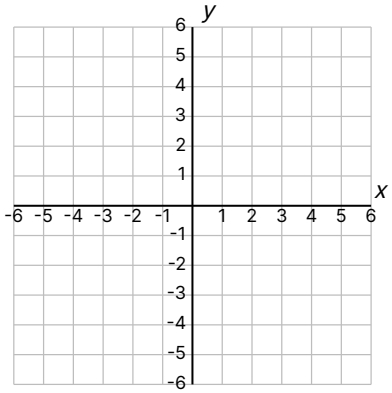
a) A: $y = x + 2$, B: $y = -x + 4$



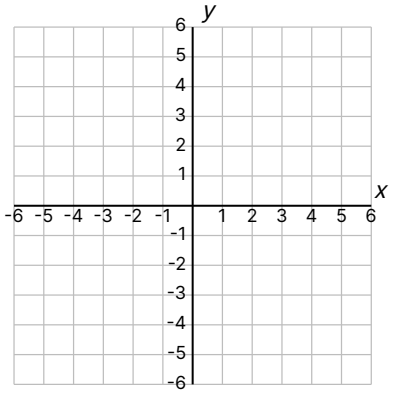
b) A: $y = 2x - 3$, B: $y = -x + 3$



c) A: $x + y = 5$, B: $y = x - 1$



d) A: $y = -2x + 1$, B: $y = x - 5$



C Elimination: add or subtract ●●○

Remember: When one letter has the same number in front of it in both equations, add or subtract the equations to remove it.

3. Solve each pair of equations.

a)
$$x + y = 10 \quad (1)$$
$$x - y = 4 \quad (2)$$

b)
$$2x + y = 9 \quad (1)$$
$$x - y = 3 \quad (2)$$

c)
$$3x + 2y = 16 \quad (1)$$
$$x + 2y = 8 \quad (2)$$

d)
$$5x + 3y = 18 \quad (1)$$
$$2x + 3y = 9 \quad (2)$$

e)
$$4x - y = 5 \quad (1)$$
$$2x + y = 7 \quad (2)$$

f)
$$3x + 2y = 16 \quad (1)$$
$$3x - y = 1 \quad (2)$$

g)
$$2x + 3y = 16 \quad (1)$$
$$2x + y = 12 \quad (2)$$

h)
$$x + 2y = 6 \quad (1)$$
$$-x + 3y = -1 \quad (2)$$

i)
$$5x - 2y = 12 \quad (1)$$
$$3x + 2y = 4 \quad (2)$$

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D Elimination: multiply first ●●●

Remember: Multiply one or both equations so that one letter has the same number in front of it. Then add or subtract.

4. Solve each pair of equations.

a) $2x + y = 6 \quad (1)$, $3x + 2y = 11 \quad (2)$

b) $3x - 2y = 4 \quad (1)$, $x + 4y = 6 \quad (2)$

c) $2x + 3y = 11 \quad (1)$, $5x - y = 2 \quad (2)$

d) $2x + 3y = 12 \quad (1)$, $3x + 2y = 13 \quad (2)$

e) $3x + 4y = 2 \quad (1)$, $2x - 3y = 7 \quad (2)$

f) $5x + 2y = 16 \quad (1)$, $3x + 4y = 18 \quad (2)$

g) $4x + 3y = 10 \quad (1)$, $2x + 5y = 12 \quad (2)$

h) $2x + 5y = 13 \quad (1)$, $3x - 2y = 10 \quad (2)$

i) $3x + 5y = 7 \quad (1)$, $4x + 3y = 2 \quad (2)$

E Substitution ●●○

Remember: Replace a letter in one equation with the expression it is equal to in the other equation.

5. Solve each pair of equations.

a) $y = 2x + 1 \quad (1)$, $3x + y = 11 \quad (2)$

b) $y = x + 3 \quad (1)$, $2x + y = 15 \quad (2)$

c) $y = 3x - 2 \quad (1)$, $x + 2y = 17 \quad (2)$

d) $x = 2y \quad (1)$, $3x + y = 14 \quad (2)$

e) $x = y + 2 \quad (1)$, $2x + 3y = 19 \quad (2)$

f) $y = 2x - 5 \quad (1)$, $3x - y = 9 \quad (2)$

g) $y = 3x - 1 \quad (1)$, $y = x + 3 \quad (2)$

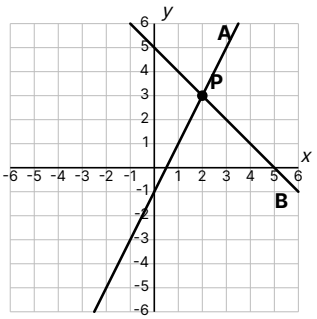
h) $y = 2x + 3 \quad (1)$, $y = -x + 6 \quad (2)$

i) $y = -2x + 7 \quad (1)$, $7x + 3y = 22 \quad (2)$

F The graph and the algebra agree ●●●

Remember: Solving by algebra and reading the graph must give the same point. If the lines are parallel they never cross, so the equations have no solution.

6. The two lines are drawn on the grid.

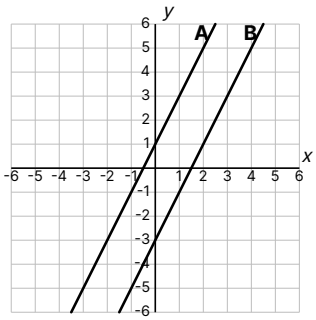


a) Solve the pair of equations by substitution: $y = 2x - 1$ and $y = -x + 5$.

b) Where is the solution on the graph? Explain why the algebra and the graph give the same answer.

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7. The lines $y = 2x + 1$ and $y = 2x - 3$ are drawn on the grid.



a) Try to solve the pair of equations by substitution. What happens?

b) Explain, using the graph, why there is no solution.

G Problems in words ●●●

8. Write two equations for each problem, then solve them.

a) Two numbers add to 20. Their difference is 6. Find the two numbers.

b) 2 adult tickets and 3 child tickets cost $45. 1 adult ticket and 2 child tickets cost $26. Find the cost of each kind of ticket.

c) 3 pies and 2 drinks cost $23. 1 pie and 4 drinks cost $21. Find the cost of a pie and the cost of a drink.

d) A rectangle is x cm wide and y cm long. Its length is 4 cm more than its width, and its perimeter is 32 cm. Find its width and its length.

e) A jar holds x 20-cent coins and y 50-cent coins. There are 14 coins altogether, worth $4.60. How many of each coin are there?

★ Extension

9. Rearrange both equations so the letters are on the left. Then solve them: $2x = 7 - y$ and $3y = x + 7$.

10. Solve $2x + 3y = 8$ and $4x - 3y = 7$. One of the answers is not a whole number.

11. The lines $y = 2x - 1$ and $y = -x + 5$ cross at one point. Find the coordinates of that point.

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Simultaneous Equations

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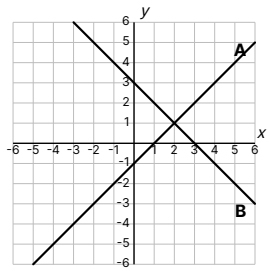
KEY WORDS simultaneous equations *(whārite pāhekoheko)* · elimination · substitution *(whakakapinga)*

A The solution on a graph ●○○

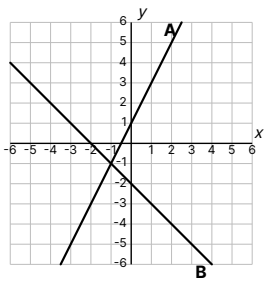
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1. Lines A and B are drawn on each grid. Write down the solution of the pair of equations.

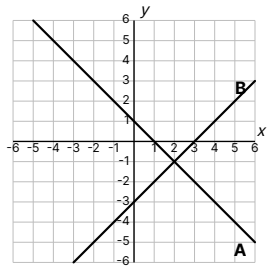
a) A: $y = x - 1$, B: $y = -x + 3$



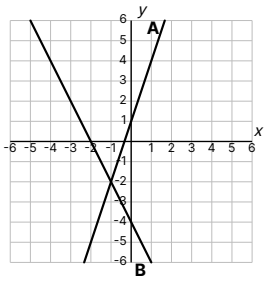
b) A: $y = 2x + 1$, B: $y = -x - 2$



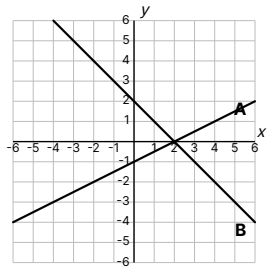
c) A: $x + y = 1$, B: $y = x - 3$



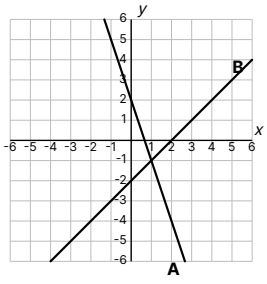
d) A: $y = 3x + 1$, B: $y = -2x - 4$



e) A: $y = \frac{1}{2}x - 1$, B: $x + y = 2$



f) A: $y = -3x + 2$, B: $y = x - 2$



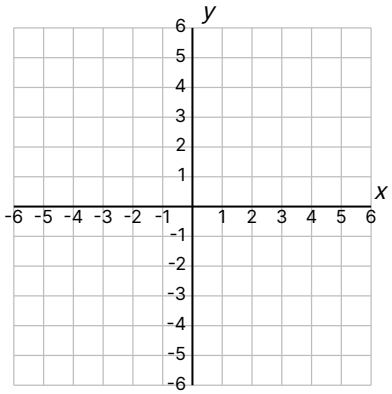
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B Solving by drawing ●●○

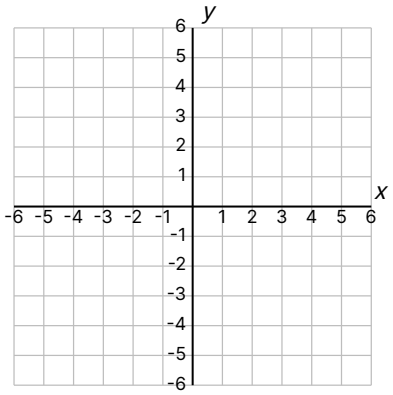
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2. Draw both lines on the grid and write down the solution.

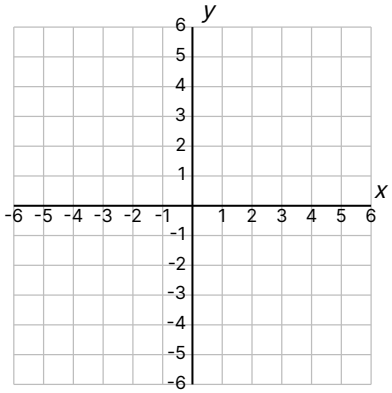
a) A: $y = x - 2$, B: $y = -x + 2$



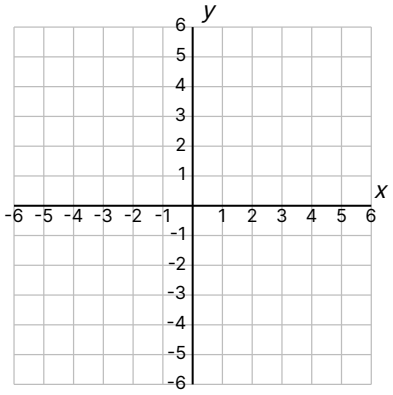
b) A: $y = 2x + 1$, B: $y = -x + 4$



c) A: $x + y = 3$, B: $y = x - 3$



d) A: $y = -2x - 1$, B: $y = x + 5$



C Elimination: add or subtract ●●○

Remember: When one letter has the same number in front of it in both equations, add or subtract the equations to remove it.

3. Solve each pair of equations.

a)
$$x + y = 13 \quad (1)$$
$$x - y = 5 \quad (2)$$

b)
$$3x + y = 11 \quad (1)$$
$$x - y = 1 \quad (2)$$

c)
$$4x + 3y = 18 \quad (1)$$
$$x + 3y = 9 \quad (2)$$

d)
$$5x + 2y = 22 \quad (1)$$
$$3x + 2y = 14 \quad (2)$$

e)
$$3x - y = 5 \quad (1)$$
$$2x + y = 10 \quad (2)$$

f)
$$2x + 3y = 18 \quad (1)$$
$$2x - y = 2 \quad (2)$$

g)
$$3x + 4y = 18 \quad (1)$$
$$3x + y = 9 \quad (2)$$

h)
$$x + 3y = 8 \quad (1)$$
$$-x + 2y = -3 \quad (2)$$

i)
$$4x - 3y = 15 \quad (1)$$
$$2x + 3y = 3 \quad (2)$$

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D Elimination: multiply first ●●●

> **Remember:** Multiply one or both equations so that one letter has the same number in front of it. Then add or subtract.

4. Solve each pair of equations.

a) $3x + y = 8 \quad (1)$
$4x + 2y = 14 \quad (2)$

b) $2x - 3y = 3 \quad (1)$
$x + 6y = 9 \quad (2)$

c) $3x + 2y = 7 \quad (1)$
$4x - y = 2 \quad (2)$

d) $3x + 2y = 12 \quad (1)$
$2x + 3y = 13 \quad (2)$

e) $4x + 3y = 1 \quad (1)$
$3x - 2y = 5 \quad (2)$

f) $7x + 2y = 16 \quad (1)$
$3x + 4y = 10 \quad (2)$

g) $2x + 3y = 8 \quad (1)$
$4x + y = 6 \quad (2)$

h) $3x + 4y = 19 \quad (1)$
$2x - 3y = 7 \quad (2)$

i) $2x + 3y = 5 \quad (1)$
$5x + 2y = -4 \quad (2)$

E Substitution ●●○

> **Remember:** Replace a letter in one equation with the expression it is equal to in the other equation.

5. Solve each pair of equations.

a) $y = 2x + 3 \quad (1)$
$2x + y = 11 \quad (2)$

b) $y = x + 5 \quad (1)$
$3x + y = 17 \quad (2)$

c) $y = 2x - 3 \quad (1)$
$x + 3y = 19 \quad (2)$

d) $x = 3y \quad (1)$
$2x + y = 14 \quad (2)$

e) $x = y + 3 \quad (1)$
$3x + 2y = 19 \quad (2)$

f) $y = 3x - 4 \quad (1)$
$5x - y = 10 \quad (2)$

g) $y = 4x - 3 \quad (1)$
$y = x + 6 \quad (2)$

h) $y = 3x + 1 \quad (1)$
$y = -x + 9 \quad (2)$

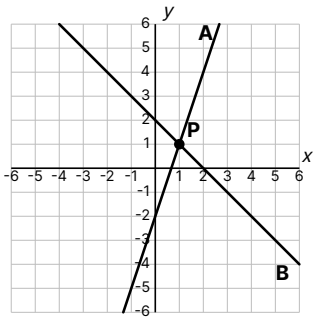
i) $y = -3x + 10 \quad (1)$
$10x + 3y = 32 \quad (2)$

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F The graph and the algebra agree ●●●

> **Remember:** Solving by algebra and reading the graph must give the same point. If the lines are parallel they never cross, so the equations have no solution.

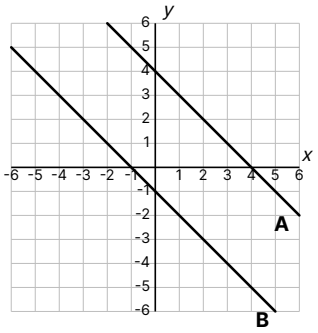
6. The two lines are drawn on the grid.



a) Solve the pair of equations by substitution: $y = 3x - 2$ and $y = -x + 2$.

b) Where is the solution on the graph? Explain why the algebra and the graph give the same answer.

7. The lines $y = -x + 4$ and $y = -x - 1$ are drawn on the grid.



a) Try to solve the pair of equations by substitution. What happens?

b) Explain, using the graph, why there is no solution.

G Problems in words ●●●

8. Write two equations for each problem, then solve them.

a) Two numbers add to 25. Their difference is 7. Find the two numbers.

b) 3 adult tickets and 2 child tickets cost \$63. 2 adult tickets and 1 child ticket cost \$39. Find the cost of each kind of ticket.

c) 2 burgers and 3 drinks cost \$27. 4 burgers and 1 drink cost \$39. Find the cost of a burger and the cost of a drink.

d) A rectangle is x cm wide and y cm long. Its length is 5 cm more than its width, and its perimeter is 38 cm. Find its width and its length.

e) A jar holds x 20-cent coins and y 50-cent coins. There are 16 coins altogether, worth \$5.30. How many of each coin are there?

★ Extension

9. Rearrange both equations so the letters are on the left. Then solve them: $3x = 11 - y$ and $2y = x + 1$.

10. Solve $2x + y = 5$ and $4x - y = 4$. One of the answers is not a whole number.

11. The lines $y = 3x - 2$ and $y = -x + 6$ cross at one point. Find the coordinates of that point.

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